AP CALCULUS AB

Unit 1 • Lesson 1.5 practice

Lesson 1.5 • Determining Limits Using Algebraic Properties of Limits

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$\lim_{x \to 2} \left(3x^2 - x\right)$ is

(A) 10

(B) 12

(C) 14

(D) 8

Question 2

NO CALCULATOR

If $\lim_{x \to c} f(x) = 4$ and $\lim_{x \to c} g(x) = -1$, then $\lim_{x \to c} \left(f(x)g(x) + 3\right)$ is

(A) 7

(B) -1

(C) 3

(D) -4

Question 3

NO CALCULATOR

If $\lim_{x \to c} f(x) = 6$ and $\lim_{x \to c} g(x) = 2$, then $\lim_{x \to c} \dfrac{f(x)}{g(x)}$ is

(A) 12

(B) 4

(C) 3

(D) $\dfrac{1}{3}$

Question 4

NO CALCULATOR

$\lim_{x \to \pi} \cos x$ is

(A) 0

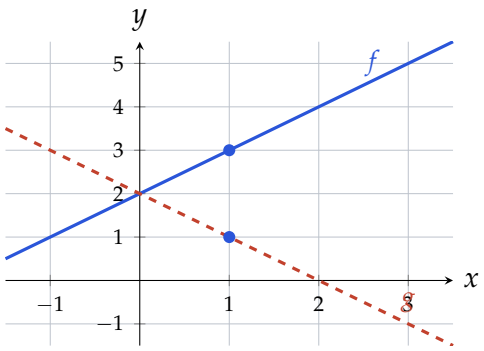
(B) 1

(C) π

(D) -1

Question 5 (graph) NO CALCULATOR

The graphs of the lines f (solid) and g (dashed) are shown. What is $\lim_{x \to 1} f(x)\, g(x)$?



(A) 4
(B) 2
(C) 1
(D) 3

Question 6 (Free response) NO CALCULATOR

$\lim_{x \to 1} f(x) = 3$, $\lim_{x \to 1} g(x) = -2$ and $h(x) = x^2 + 1$. Find each limit, showing which properties you use.

(a) $\lim_{x \to 1} \left(2f(x) - g(x)\right)$ (b) $\lim_{x \to 1} f(x)h(x)$

(c) $\lim_{x \to 1} \sqrt{f(x) + 6}$ (d) $\lim_{x \to 1} \left(g(x)\right)^3$